# PROMETHEUS Patent Claims

Hybrid Analog–Digital Computing System with Time-Domain Multiplexed Computation

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# 1 PATENT CLAIMS

## 1.1 Claim 1 (Independent Claim)

A hybrid analog–digital computing system, comprising:

a) an analog computation fabric configured to perform continuous-time mathematical operations;

b) a digital control subsystem configured to control, schedule, and supervise operation of the analog computation fabric;

c) a time-domain multiplexing mechanism by which the analog computation fabric is shared among a plurality of logical computation channels by allocating discrete time slots;

d) a calibration subsystem configured to measure and compensate for non-idealities of the analog computation fabric; and

e) a host interface enabling configuration and control of the system by a digital computing device,

wherein computation within each allocated time slot is performed in the analog domain as a continuous physical process.

## 1.2 Dependent Claims

**Claim 2.** The system of claim 1, wherein the analog computation fabric comprises one or more analog integrators configured to represent system state variables.

**Claim 3.** The system of claim 1, wherein the time-domain multiplexing mechanism is configured such that analog settling occurs within each time slot prior to evaluation of computation results.

**Claim 4.** The system of claim 1, wherein the digital control subsystem is implemented using programmable logic.

**Claim 5.** The system of claim 1, wherein the calibration subsystem applies digitally stored correction coefficients to compensate for gain error, offset, temperature drift, or component mismatch.

**Claim 6.** The system of claim 1, further comprising analog-to-digital converters and digital-to-analog converters operatively coupled to the analog computation fabric for calibration, monitoring, or configuration purposes.

**Claim 7.** The system of claim 1, wherein the plurality of logical computation channels is greater than the number of physical analog computation elements.

**Claim 8.** The system of claim 1, wherein the host interface comprises a high-speed peripheral interconnect enabling direct memory access between the system and a host processor.

**Claim 9.** The system of claim 1, wherein the system is configured as a hardware accelerator card installable in a computing system.

**Claim 10.** The system of claim 1, wherein the system is configured to execute differential equation solving, optimization, control, or signal processing tasks.